

Unlocking the Secrets of Public Feedback

Hidden Action and Information Complementarity

Research Notes

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Abstract

An agent privately chooses one of finitely many initial actions, thereby changing the distribution of a payoff-relevant state. A principal observes the realized state and sends public feedback according to an experiment committed to in advance. The agent then combines that feedback with his private action history when making a continuation decision. A single public message therefore generates a collection of history-dependent posteriors. We show that implementing any target action is a persuasion problem with one Bayes-plausibility condition and one incentive constraint for every deviation. Despite these additional constraints, the problem retains a concavification representation, now with a vector of incentive multipliers. The central economic force is *information complementarity*: the principal rewards a desirable private action by making public information more valuable when combined with the private history generated by that action. Under monotone likelihood ratios and single-crossing conditions, the general problem admits monotone comparative statics and a likelihood-ratio ordering of distortions. In a spatial model with binary continuation, the optimal policy may pool extreme low states with relatively high states. This disconnected pooling is not conventional persuasion of a common receiver; it selectively contaminates the information available after an undesirable private action.

1 Motivation

In standard Bayesian persuasion, the principal pools states in order to move a *common* posterior across an action threshold. Here the receiver of information is also the agent who previously chose a hidden action. Because he remembers that action, there is no common posterior following a public message.

The principal instead designs how public information interacts with privately known experience. The same message is interpreted through different endogenous private priors, one for each possible action history. The principal pools states in order to make the message differentially informative across those histories:

the same public feedback can be precise after effort and ambiguous after shirking.

The principal thereby provides an *informational reward* for a desirable action. Public feedback complements the private experience generated by the agent's own behavior.

The contribution has three layers. First, we formulate the problem for an arbitrary finite initial-action set and show that the familiar concavification method survives: multiple hidden-action deviations simply generate multiple moment constraints and a vector of shadow prices. Second, we seek economically interpretable conditions—monotone likelihood ratios, single crossing, and increasing differences—under which the binding deviations and optimal posterior splits vary monotonically with primitives. Third, we return to a spatial environment with a binary continuation

action, where the optimal signal can be characterized sharply and may join extreme low states with relatively high states in a disconnected pool.

A simple example illustrates the idea. Suppose

$$a = 1 \implies x \in \{2, 3\}, \quad a = 0 \implies x \in \{1, 2\}.$$

The principal reports whether or not $x = 3$, using the partition

$$\{1, 2\} \mid \{3\}.$$

After effort, the message “ $x \neq 3$ ” identifies $x = 2$. After shirking, the same message does not distinguish $x = 1$ from $x = 2$. Thus, the public signal is fully informative for the worker and uninformative for the shirker.

2 General environment

There are two periods and a finite state space

$$X = \{x_1, \dots, x_n\}.$$

At date 0, the principal commits to an experiment

$$\sigma(m \mid x),$$

where m belongs to a finite message set M .

At date 1, the agent privately chooses an action from a finite set

$$a \in A = \{1, \dots, K\}.$$

Action a has cost $k(a)$. Conditional on a , the state is drawn from $f_a \in \Delta(X)$. Unless otherwise noted, assume full support:

$$f_a(x) > 0 \quad \text{for every } x \in X \text{ and } a \in A.$$

The agent knows his own action but does not observe x . The principal observes x and sends m according to σ .

At date 2, the agent chooses $b \in B$. Payoffs are

$$U_A = u(b, x) - k(a),$$

$$U_P = R(x) + w(b, x).$$

The function R captures the principal’s direct value from the state and therefore her preference over initial actions. The function w captures the principal’s interest in the continuation decision.

Given action a and message m , the agent’s posterior is

$$\mu_a^m(x) = \frac{f_a(x)\sigma(m \mid x)}{\sum_{y \in X} f_a(y)\sigma(m \mid y)}.$$

The same message generally induces a different posterior for every possible private action history.

3 The linked-posterior representation

Fix a target action a^* . Use f_{a^*} as the on-path prior. For every deviation $a \neq a^*$, define

$$\ell_a(x) := \frac{f_a(x)}{f_{a^*}(x)}.$$

For any $\mu \in \Delta(X)$, define

$$\rho_a(\mu) := \sum_{x \in X} \ell_a(x) \mu(x)$$

and the likelihood-ratio tilt

$$T_a \mu(x) := \frac{\ell_a(x) \mu(x)}{\rho_a(\mu)}.$$

Proposition 1 (Linked posteriors). *Suppose a message induces posterior μ under a^* . For every $a \neq a^*$:*

- (i) *its probability under a equals its probability under a^* multiplied by $\rho_a(\mu)$;*
- (ii) *the posterior induced by the same message under a is $T_a \mu$.*

Proof. Let $p_a(m) = \sum_x f_a(x) \sigma(m | x)$. If $\mu(x) = f_{a^*}(x) \sigma(m | x) / p_{a^*}(m)$, then

$$\frac{p_a(m)}{p_{a^*}(m)} = \sum_x \frac{f_a(x)}{f_{a^*}(x)} \mu(x) = \rho_a(\mu).$$

Moreover,

$$\mu_a^m(x) = \frac{f_a(x) \sigma(m | x)}{p_a(m)} = \frac{\ell_a(x) \mu(x)}{\rho_a(\mu)} = T_a \mu(x).$$

□

Thus, the principal's basic choice variable is a linked collection

$$\boxed{(\mu, (T_a \mu)_{a \neq a^*})}.$$

The mappings T_a capture how a common public message is decoded under every private action history. The paired-posterior formulation is the special case $|A| = 2$.

4 Continuation values and incentive compatibility

Define the agent's value of making the optimal continuation decision under posterior μ :

$$v_A(\mu) := \max_{b \in B} \mathbb{E}_\mu[u(b, x)].$$

When the maximizer is unique, write it as $b^*(\mu)$ and define the principal's continuation value

$$v_P(\mu) := \mathbb{E}_\mu[w(b^*(\mu), x)].$$

Let τ denote the distribution of posteriors induced under the on-path prior f_{a^*} . Bayes plausibility requires

$$\mathbb{E}_\tau[\mu] = f_{a^*}.$$

The target action's continuation value is

$$V_{a^*}(\tau) = \mathbb{E}_\tau[v_A(\mu)].$$

For any deviation $a \neq a^*$, the linked-posterior representation gives

$$V_a(\tau) = \mathbb{E}_\tau[\rho_a(\mu)v_A(T_a\mu)].$$

Define the incentive contribution against deviation a by

$$h_a(\mu) := v_A(\mu) - \rho_a(\mu)v_A(T_a\mu).$$

The target action is incentive compatible if and only if, for every $a \neq a^*$,

$$\mathbb{E}_\tau[h_a(\mu)] \geq d_a, \quad d_a := k(a^*) - k(a).$$

Conditional on implementing a^* , the principal solves

$$\begin{aligned} \max_{\tau} \quad & \mathbb{E}_\tau[v_P(\mu)] \\ \text{s.t.} \quad & \mathbb{E}_\tau[\mu] = f_{a^*}, \\ & \mathbb{E}_\tau[h_a(\mu)] \geq d_a \quad \text{for every } a \neq a^*. \end{aligned} \tag{1}$$

The expected state payoff $\mathbb{E}_{f_{a^*}}[R(x)]$ is constant in this conditional problem.

4.1 Regularity

The following two assumptions are maintained whenever (1) is invoked. The first is a tie-breaking and boundedness condition; the second is a Slater condition.

Assumption 1 (Primitives and tie-breaking). (i) X , A and B are finite and $u, w : B \times X \rightarrow \mathbb{R}$ are real valued.

(ii) $f_a(x) > 0$ for every $a \in A$ and $x \in X$.

(iii) Continuation ties are resolved in the principal's favour: writing $B^*(\mu) := \arg \max_{b \in B} \mathbb{E}_\mu[u(b, x)]$, the continuation rule satisfies $b^*(\mu) \in \arg \max_{b \in B^*(\mu)} \mathbb{E}_\mu[w(b, x)]$, so that

$$v_P(\mu) = \max_{b \in B^*(\mu)} \mathbb{E}_\mu[w(b, x)].$$

(iv) First-period ties are resolved in favour of a^* , which is why the incentive constraints in (1) are weak.

Assumption 2 (Slater). There exists $\tau_0 \in \Delta(\Delta(X))$ with $\mathbb{E}_{\tau_0}[\mu] = f_{a^*}$ and $\mathbb{E}_{\tau_0}[h_a(\mu)] > d_a$ for every $a \neq a^*$.

Assumption 2 rules out the knife-edge case in which a^* is implementable but only with all incentive constraints exactly binding at every feasible experiment; Section 8 computes the exact implementability frontier in a leading example, and the frontier is precisely where Assumption 2 fails.

Lemma 1 (Structure of the objects in (1)). *Under Assumption 1, for every $a \neq a^*$:*

(i) ρ_a is linear on $\Delta(X)$ and $\min_x \ell_a(x) \leq \rho_a(\mu) \leq \max_x \ell_a(x)$, with $\min_x \ell_a(x) > 0$; consequently $\mu \mapsto T_a \mu$ is continuous on $\Delta(X)$;

(ii) for every $\mu \in \Delta(X)$,

$$\rho_a(\mu) v_A(T_a \mu) = \max_{b \in B} \mathbb{E}_\mu[\ell_a(x) u(b, x)], \quad (2)$$

so that

$$h_a(\mu) = \max_{b \in B} \mathbb{E}_\mu[u(b, x)] - \max_{b \in B} \mathbb{E}_\mu[\ell_a(x) u(b, x)] \quad (3)$$

is a difference of two convex, piecewise-linear, Lipschitz functions on $\Delta(X)$; in particular h_a is continuous and bounded;

(iii) v_P is bounded and upper semicontinuous on $\Delta(X)$.

Proof. (i) Linearity of ρ_a is immediate from its definition, and the bounds follow because $\rho_a(\mu)$ is an average of the numbers $\ell_a(x)$ with weights $\mu(x)$. Assumption 1(ii) gives $\ell_a(x) \in (0, \infty)$ for every x , so ρ_a is bounded away from 0 on $\Delta(X)$ and $T_a \mu(x) = \ell_a(x) \mu(x) / \rho_a(\mu)$ is a ratio of continuous functions with non-vanishing denominator.

(ii) For any $b \in B$,

$$\rho_a(\mu) \sum_x T_a \mu(x) u(b, x) = \rho_a(\mu) \sum_x \frac{\ell_a(x) \mu(x)}{\rho_a(\mu)} u(b, x) = \sum_x \ell_a(x) \mu(x) u(b, x).$$

Taking the maximum over the finite set B and using $\rho_a(\mu) > 0$ gives (2); substituting into the definition of h_a gives (3). Each of the two terms in (3) is a maximum of finitely many linear functions of μ , hence convex, piecewise linear and Lipschitz on the compact set $\Delta(X)$.

(iii) Boundedness follows from finiteness of $B \times X$. For upper semicontinuity, let $\mu_k \rightarrow \mu$ and pass to a subsequence along which $v_P(\mu_k) \rightarrow \limsup_k v_P(\mu_k)$ and, using finiteness of B , along which the selected continuation action is a fixed $b \in B^*(\mu_k)$ for all k . For every $b' \in B$, $\mathbb{E}_{\mu_k}[u(b, x)] \geq \mathbb{E}_{\mu_k}[u(b', x)]$; passing to the limit gives $b \in B^*(\mu)$. Hence $\limsup_k v_P(\mu_k) = \lim_k \mathbb{E}_{\mu_k}[w(b, x)] = \mathbb{E}_\mu[w(b, x)] \leq v_P(\mu)$. $\square$

Proposition 2 (Constrained concavification). *For a multiplier vector $\lambda = (\lambda_a)_{a \neq a^*} \geq 0$, define*

$$G_\lambda(\mu) := v_P(\mu) + \sum_{a \neq a^*} \lambda_a h_a(\mu).$$

Under the standard finite-state regularity conditions, the value of (1) is

$$W(a^*) = \min_{\lambda \in \mathbb{R}_+^{K-1}} \left\{ \text{cav } G_\lambda(f_{a^*}) - \sum_{a \neq a^*} \lambda_a d_a \right\}.$$

The minimum is attained. Moreover, there is an optimal experiment with at most $n + K - 1$ posterior realizations, and if exactly r incentive constraints bind at the selected optimum the bound tightens to $n + r$.

Proof. The proof proceeds in four steps: (i) existence of an optimum, (ii) weak duality together with the concavification identity, (iii) strong duality and attainment of the dual minimum, (iv) the support bound.

Step 1 (Existence). Let

$$\mathcal{T} := \{\tau \in \Delta(\Delta(X)) : \mathbb{E}_\tau[\mu] = f_{a^*}\}, \quad \mathcal{T}(d) := \{\tau \in \mathcal{T} : \mathbb{E}_\tau[h_a(\mu)] \geq d_a \ \forall a \neq a^*\},$$

where $\Delta(\Delta(X))$ carries the weak-* topology. Since X is finite, $\Delta(X)$ is a compact metric space, so $\Delta(\Delta(X))$ is weak-* compact and metrizable. The maps $\tau \mapsto \mathbb{E}_\tau[\mu(x)]$ are weak-* continuous because each $\mu \mapsto \mu(x)$ is continuous and bounded, so $\mathcal{T}$ is closed; by Lemma 1(ii) each h_a is continuous and bounded, so $\mathcal{T}(d)$ is closed. Both sets are convex, being defined by linear restrictions on τ . Hence $\mathcal{T}(d)$ is convex and weak-* compact, and it is non-empty by Assumption 2.

By Lemma 1(iii), v_P is bounded and upper semicontinuous, so by the portmanteau theorem $\tau \mapsto \mathbb{E}_\tau[v_P(\mu)]$ is upper semicontinuous on $\Delta(\Delta(X))$; it is also linear. An upper semicontinuous function on a non-empty compact set attains its maximum, so (1) has a solution. Write

$$\Phi(d') := \max_{\tau \in \mathcal{T}(d')} \mathbb{E}_\tau[v_P(\mu)] \in [-\infty, \infty), \quad d' \in \mathbb{R}^{K-1},$$

with $\Phi(d') := -\infty$ when $\mathcal{T}(d') = \emptyset$, so that $W(a^*) = \Phi(d)$.

Step 2 (Weak duality and the concavification identity). Fix $\lambda \in \mathbb{R}_+^{K-1}$. First,

$$\sup_{\tau \in \mathcal{T}} \mathbb{E}_\tau[G_\lambda(\mu)] = \text{cav } G_\lambda(f_{a^*}). \quad (4)$$

For “ $\geq$ ”: by definition $\text{cav } G_\lambda(f_{a^*})$ is the supremum of $\sum_i \theta_i G_\lambda(\mu_i)$ over finite collections with $\sum_i \theta_i \mu_i = f_{a^*}$, and every such collection defines an element of $\mathcal{T}$. For “ $\leq$ ”: $\text{cav } G_\lambda$ is concave and bounded on $\Delta(X)$, hence Borel measurable, and $G_\lambda \leq \text{cav } G_\lambda$ pointwise, so Jensen’s inequality gives

$$\mathbb{E}_\tau[G_\lambda(\mu)] \leq \mathbb{E}_\tau[\text{cav } G_\lambda(\mu)] \leq \text{cav } G_\lambda(\mathbb{E}_\tau[\mu]) = \text{cav } G_\lambda(f_{a^*})$$

for every $\tau \in \mathcal{T}$. Second, for $\tau \in \mathcal{T}(d)$ and $\lambda \geq 0$,

$$\mathbb{E}_\tau[v_P(\mu)] \leq \mathbb{E}_\tau[v_P(\mu)] + \sum_{a \neq a^*} \lambda_a (\mathbb{E}_\tau[h_a(\mu)] - d_a) = \mathbb{E}_\tau[G_\lambda(\mu)] - \sum_{a \neq a^*} \lambda_a d_a.$$

Combining with (4) and taking the infimum over $\lambda \geq 0$,

$$\Phi(d) \leq \inf_{\lambda \in \mathbb{R}_+^{K-1}} \left\{ \text{cav } G_\lambda(f_{a^*}) - \sum_{a \neq a^*} \lambda_a d_a \right\}. \quad (5)$$

Step 3 (Strong duality). Φ is concave on $\mathbb{R}^{K-1}$: if $\tau^1 \in \mathcal{T}(d^1)$ and $\tau^2 \in \mathcal{T}(d^2)$ and $\theta \in [0, 1]$, then $\theta\tau^1 + (1-\theta)\tau^2 \in \mathcal{T}(\theta d^1 + (1-\theta)d^2)$, because both the Bayes-plausibility restriction and the incentive restrictions are linear in τ ; the objective is linear in τ . Φ is also non-increasing, since $d' \leq d''$ implies $\mathcal{T}(d'') \subseteq \mathcal{T}(d')$.

By Assumption 2 there is $\tau_0 \in \mathcal{T}$ with $d_a^0 := \mathbb{E}_{\tau_0}[h_a(\mu)] > d_a$ for all a . For any d' with $d' \leq d^0$ we have $\tau_0 \in \mathcal{T}(d')$, so $\Phi(d') > -\infty$. Hence the open set $\{d' : d' < d^0\}$ is contained in $\text{dom } \Phi$ and contains d , so $d \in \text{int dom } \Phi$. A concave function is superdifferentiable on the interior of its domain, so there exists $g \in \mathbb{R}^{K-1}$ with

$$\Phi(d') \leq \Phi(d) + g \cdot (d' - d) \quad \text{for all } d' \in \mathbb{R}^{K-1}. \quad (6)$$

Because Φ is non-increasing, taking $d' = d + te_a$ with $t > 0$ in (6) gives $g_a \leq 0$; set $\lambda^* := -g \geq 0$.

Now take any $\tau \in \mathcal{T}$ and put $d'_a := \mathbb{E}_\tau[h_a(\mu)]$, so that $\tau \in \mathcal{T}(d')$ and $\mathbb{E}_\tau[v_P(\mu)] \leq \Phi(d')$. By (6),

$$\mathbb{E}_\tau[v_P(\mu)] \leq \Phi(d) - \sum_{a \neq a^*} \lambda_a^* (\mathbb{E}_\tau[h_a(\mu)] - d_a),$$

that is, $\mathbb{E}_\tau[G_{\lambda^*}(\mu)] - \sum_a \lambda_a^* d_a \leq \Phi(d)$. Taking the supremum over $\tau \in \mathcal{T}$ and using (4),

$$\text{cav } G_{\lambda^*}(f_{a^*}) - \sum_{a \neq a^*} \lambda_a^* d_a \leq \Phi(d).$$

Together with (5) this yields equality and shows that the infimum is attained at λ^* .

Step 4 (Support bound). $\mathcal{T}(d)$ is a non-empty compact convex subset of the locally convex space of finite signed Borel measures on $\Delta(X)$ with the weak-* topology, and $\tau \mapsto \mathbb{E}_\tau[v_P(\mu)]$ is upper semicontinuous and convex, being linear. By Bauer's maximum principle it attains its maximum at an extreme point τ^* of $\mathcal{T}(d)$. Let $\mathcal{B} := \{a \neq a^* : \mathbb{E}_{\tau^*}[h_a(\mu)] = d_a\}$ and $r := |\mathcal{B}| \leq K - 1$.

Suppose $\text{supp } \tau^*$ contains at least $n+r+1$ distinct points $\mu_1, \dots, \mu_{n+r+1}$. Choose pairwise disjoint open sets $O_1, \dots, O_{n+r+1}$ with $\mu_i \in O_i$; then $\pi_i := \tau^*(O_i) > 0$ for each i , and $\nu_i := \tau^*(\cdot \cap O_i)/\pi_i$ is a probability measure on $\Delta(X)$. Consider the linear system in $\epsilon \in \mathbb{R}^{n+r+1}$

$$\sum_i \epsilon_i = 0, \quad \sum_i \epsilon_i \mathbb{E}_{\nu_i}[\mu(x)] = 0 \quad (x \in X), \quad \sum_i \epsilon_i \mathbb{E}_{\nu_i}[h_a(\mu)] = 0 \quad (a \in \mathcal{B}).$$

Summing the n middle equations reproduces the first, so at most $1 + (n-1) + r = n+r$ of these equations are independent. Since the system has $n+r+1$ unknowns, it admits a solution $\epsilon \neq 0$.

Define $\tau^\pm := \tau^* \pm t \sum_i \epsilon_i \nu_i$ for $t > 0$. Because the O_i are disjoint and $t|\epsilon_i|\nu_i \leq \pi_i^{-1}t|\epsilon_i|\tau^*(\cdot \cap O_i)$, choosing $t \leq \min_i \pi_i/|\epsilon_i|$ (over i with $\epsilon_i \neq 0$) makes $\tau^\pm$ non-negative. Their total mass is $1 \pm t \sum_i \epsilon_i = 1$; they satisfy $\mathbb{E}_{\tau^\pm}[\mu] = f_{a^*}$ and $\mathbb{E}_{\tau^\pm}[h_a(\mu)] = d_a$ for $a \in \mathcal{B}$; and for $a \notin \mathcal{B}$ the constraint holds strictly at τ^* , hence continues to hold for t small. Thus $\tau^\pm \in \mathcal{T}(d)$ for small $t > 0$, $\tau^+ \neq \tau^-$, and $\tau^* = \frac{1}{2}(\tau^+ + \tau^-)$, contradicting extremality. Hence $|\text{supp } \tau^*| \leq n+r \leq n+K-1$.

Finally, a finitely supported τ^* with $\mathbb{E}_{\tau^*}[\mu] = f_{a^*}$ is induced by an explicit experiment with message set $M = \text{supp } \tau^*$ and $\sigma(\mu \mid x) := \tau^*(\mu)\mu(x)/f_{a^*}(x)$, which is well defined by Assumption 1(ii), satisfies $\sum_{\mu \in M} \sigma(\mu \mid x) = 1$, generates $m = \mu$ with probability $\tau^*(\mu)$ under f_{a^*} , and induces posterior μ after message μ . $\square$

Remark 1 (Where tie-breaking matters). *Assumption 1(iii) is used only in Lemma 1(iii) and hence only for the existence claim in Step 1 and for Bauer's principle in Step 4. Under the opposite tie-breaking convention v_P is lower semicontinuous rather than upper semicontinuous, the maximum in (1) need not be attained, and $W(a^*)$ should be read as a supremum approached by experiments that perturb posteriors away from the indifference set. The duality in Steps 2–3 does not use the convention: it relies only on continuity of h_a , which holds by Lemma 1(ii) regardless of how ties are resolved.*

The result separates geometry from incentives. Bayes plausibility is unchanged from standard persuasion; private action histories add linear moment inequalities. Each multiplier λ_a is the shadow value of deterring one deviation. This formulation is useful even when no closed-form signal exists, because the target-action problem remains a finite-dimensional constrained concavification.

5 Information complementarity

For any experiment σ , define its value under private history a by

$$I_a(\sigma) := V_a(\sigma) - v_A(f_a).$$

This is the value of public information when combined with the private prior generated by action a . For every deviation $a \neq a^*$, the gross incentive wedge decomposes as

$$V_{a^*}(\sigma) - V_a(\sigma) = \underbrace{v_A(f_{a^*}) - v_A(f_a)}_{\text{value of changing the private prior}} + \underbrace{I_{a^*}(\sigma) - I_a(\sigma)}_{\text{information-complementarity wedge}}. \quad (7)$$

The purest version of the mechanism imposes

$$v_A(f_{a^*}) = v_A(f_a)$$

for the relevant deviations, so that all incentives arise from

$$I_{a^*}(\sigma) - I_a(\sigma).$$

The principal then rewards the target action solely by designing a public signal that is more useful when combined with the history generated by that action.

A natural comparative notion is increasing differences between private history and public information. If σ' is more informative than σ , information and effort are complementary when

$$V_{a^*}(\sigma') - V_{a^*}(\sigma) \geq V_a(\sigma') - V_a(\sigma).$$

The principal's design problem is not necessarily to maximize overall informativeness, but to choose an experiment for which this inequality is strong.

6 A solvable specialization: binary continuation action

Return now to two initial actions, $A = \{0, 1\}$, with $k(1) = c$, $k(0) = 0$, and target $a^* = 1$. Write

$$\ell(x) := \frac{f_0(x)}{f_1(x)}, \quad \rho(\mu) := \mathbb{E}_\mu[\ell(x)], \quad T\mu(x) := \frac{\ell(x)\mu(x)}{\rho(\mu)}.$$

Let

$$B = \{0, 1\}$$

and suppose the principal and agent have aligned continuation preferences:

$$u(0, x) = w(0, x) = 0, \quad u(1, x) = w(1, x) = q(x).$$

The continuation action $b = 1$ can be interpreted as continuing the project or exerting additional effort. State x warrants continuation if $q(x) > 0$ and stopping if $q(x) < 0$.

The agent's continuation value is

$$v(\mu) = [\mathbb{E}_\mu q(x)]^+,$$

where $[z]^+ := \max\{z, 0\}$. Since

$$\rho(\mu)\mathbb{E}_{T\mu}q(x) = \mathbb{E}_\mu[\ell(x)q(x)],$$

the incentive contribution becomes

$$h(\mu) = [\mathbb{E}_\mu q]^+ - [\mathbb{E}_\mu(\ell q)]^+.$$

Consequently,

$$G_\lambda(\mu) = (1 + \lambda) [\mathbb{E}_\mu q]^+ - \lambda [\mathbb{E}_\mu(\ell q)]^+.$$

This function is piecewise linear on the posterior simplex, so the general problem is a finite polyhedral concavification.

6.1 Two messages suffice

Proposition 3 (Recommendation reduction). *For every experiment σ there is a two-message experiment $\hat{\sigma}$, with messages*

$$W : b = 1, \quad S : b = 0,$$

such that the principal's on-path payoff is unchanged, $V_1(\hat{\sigma}) = V_1(\sigma)$ and $V_0(\hat{\sigma}) \leq V_0(\sigma)$. Consequently the value of the target-effort problem is unchanged by restricting to two messages, and it is attained.

Proof. Let M be the message set of σ , let $p_i(m) := \sum_x f_i(x)\sigma(m | x)$ for $i \in \{0, 1\}$, and let μ^m denote the worker's posterior after m . Write

$$Q_i^m := \sum_x f_i(x)\sigma(m | x)q(x),$$

so that $Q_1^m = p_1(m)\mathbb{E}_{\mu^m}[q]$ and $V_i(\sigma) = \sum_{m \in M} [Q_i^m]^+$ by the form of v in the display preceding this subsection.

Partition $M = M_W \cup M_S$ with $M_W := \{m \in M : Q_1^m \geq 0\}$, ties assigned to M_W , and set $\hat{\sigma}(W | x) := \sum_{m \in M_W} \sigma(m | x)$ and $\hat{\sigma}(S | x) := 1 - \hat{\sigma}(W | x)$.

Worker. $\hat{Q}_1^W = \sum_{m \in M_W} Q_1^m \geq 0$ and $\hat{Q}_1^S = \sum_{m \in M_S} Q_1^m \leq 0$, so under $\hat{\sigma}$ the worker indeed chooses $b = 1$ after W and $b = 0$ after S , and

$$V_1(\hat{\sigma}) = [\hat{Q}_1^W]^+ + [\hat{Q}_1^S]^+ = \sum_{m \in M_W} Q_1^m = \sum_{m \in M} [Q_1^m]^+ = V_1(\sigma).$$

Since $w(b, \cdot) = u(b, \cdot)$ for both b , the principal's on-path continuation payoff equals V_1 and is likewise unchanged.

Shirker. By subadditivity of $z \mapsto [z]^+$,

$$V_0(\hat{\sigma}) = \left[\sum_{m \in M_W} Q_0^m \right]^+ + \left[\sum_{m \in M_S} Q_0^m \right]^+ \leq \sum_{m \in M} [Q_0^m]^+ = V_0(\sigma).$$

Hence $V_1(\hat{\sigma}) - V_0(\hat{\sigma}) \geq V_1(\sigma) - V_0(\sigma)$, so if σ satisfies the first-period incentive constraint then so does $\hat{\sigma}$.

Attainment. A two-message experiment is a vector $\alpha \in [0, 1]^n$ with $\alpha_x = \mathbb{P}(W | x)$. The maps $\alpha \mapsto Q_i^W$, $\alpha \mapsto Q_i^S$ are linear and $\alpha \mapsto V_0 = [Q_0^W]^+ + [Q_0^S]^+$ is continuous, so the feasible set is a closed subset of the compact set $[0, 1]^n$ and the continuous objective attains its maximum whenever the feasible set is non-empty. $\square$

Let

$$\alpha_x := \mathbb{P}(W | x).$$

For $i \in \{0, 1\}$, define

$$Q_i^W := \sum_x f_i(x)\alpha_x q(x), \quad Q_i^S := \sum_x f_i(x)(1 - \alpha_x)q(x).$$

On-path obedience requires

$$Q_1^W \geq 0, \quad Q_1^S \leq 0.$$

Introduce z_W, z_S to represent the shirker's optimized payoff after each message. The target-effort problem is the linear program

$$\begin{aligned}
& \max_{\alpha, z_W, z_S} && Q_1^W \\
& \text{s.t.} && Q_1^W - z_W - z_S \geq c, \\
& && Q_1^W \geq 0, \quad Q_1^S \leq 0, \\
& && z_W \geq Q_0^W, \quad z_W \geq 0, \\
& && z_S \geq Q_0^S, \quad z_S \geq 0, \\
& && 0 \leq \alpha_x \leq 1 \quad \text{for all } x.
\end{aligned} \tag{8}$$

Lemma 2 (The linearisation is exact). *Program (8) is equivalent to the two-message target-effort problem: at every optimum of (8), $z_W = [Q_0^W]^+$ and $z_S = [Q_0^S]^+$, so the first constraint of (8) is exactly $V_1 - V_0 \geq c$.*

Proof. The variables z_W, z_S do not enter the objective and appear in the constraint set only through the two lower bounds $z_W \geq \max\{Q_0^W, 0\}$, $z_S \geq \max\{Q_0^S, 0\}$ and through $-z_W - z_S$ in the incentive constraint, where they enter with a negative coefficient. Hence, given any feasible (α, z_W, z_S) , replacing z_W by $[Q_0^W]^+$ and z_S by $[Q_0^S]^+$ preserves feasibility, weakly relaxes the incentive constraint, and leaves the objective unchanged. Substituting these values, the incentive constraint reads $Q_1^W - [Q_0^W]^+ - [Q_0^S]^+ \geq c$, which is $V_1 - V_0 \geq c$ given on-path obedience. $\square$

Thus, the finite-state model is globally solvable even outside the region admitting a closed-form ranking result.

7 Likelihood-ratio characterization in the regular region

Suppose both the worker and the shirker follow the recommendations:

$$Q_i^W \geq 0, \quad Q_i^S \leq 0, \quad i \in \{0, 1\}.$$

Under full information about the sign of q , set

$$\alpha_x^{FI} = \begin{cases} 1, & q(x) > 0, \\ 0, & q(x) < 0. \end{cases}$$

The corresponding incentive wedge is

$$\Delta^{FI} := \sum_{q(x) > 0} [f_1(x) - f_0(x)] q(x).$$

If $c \leq \Delta^{FI}$, the efficient continuation policy already induces first-period effort.

Suppose $c > \Delta^{FI}$. Define the distortion probability

$$\delta_x = \begin{cases} 1 - \alpha_x, & q(x) > 0, \\ \alpha_x, & q(x) < 0. \end{cases}$$

The worker's continuation payoff is

$$V_1 = V_1^{FI} - \sum_x f_1(x) |q(x)| \delta_x,$$

while the incentive wedge is

$$V_1 - V_0 = \Delta^{FI} + \sum_x [f_0(x) - f_1(x)] |q(x)| \delta_x.$$

The principal therefore solves the fractional knapsack problem

$$\begin{aligned} \min_{\delta} \quad & \sum_x f_1(x) |q(x)| \delta_x \\ \text{s.t.} \quad & \sum_x [f_0(x) - f_1(x)] |q(x)| \delta_x \geq c - \Delta^{FI}, \\ & 0 \leq \delta_x \leq 1. \end{aligned} \tag{9}$$

Throughout this section assume $q(x) \neq 0$ for every x , so that α^{FI} is well defined and the per-state cost $f_1(x)|q(x)|$ is strictly positive.

Proposition 4 (Likelihood-ratio ranking). *Program (9) is feasible if and only if*

$$\sum_{x: \ell(x) > 1} [f_0(x) - f_1(x)] |q(x)| \geq c - \Delta^{FI}, \tag{10}$$

and in that case it has a solution δ^* with the following structure. Order the states with $\ell(x) > 1$ as $y_1, y_2, \dots$ so that $\ell(y_1) \geq \ell(y_2) \geq \dots$, write $b_{y_k} := [f_0(y_k) - f_1(y_k)] |q(y_k)| > 0$, and let $j := \min\{k : \sum_{i \leq k} b_{y_i} \geq c - \Delta^{FI}\}$. Then

$$\delta_{y_k}^* = 1 \ (k < j), \quad \delta_{y_j}^* = \frac{c - \Delta^{FI} - \sum_{i < j} b_{y_i}}{b_{y_j}} \in (0, 1], \quad \delta_x^* = 0 \text{ otherwise.}$$

In particular, states are distorted in descending order of $\ell(x) = f_0(x)/f_1(x)$, the constraint binds, and at most one state is randomized. If ℓ is injective on $\{x : \ell(x) > 1\}$ this is the unique solution.

Proof. Write $c_x := f_1(x)|q(x)| > 0$ and $b_x := [f_0(x) - f_1(x)] |q(x)| = (\ell(x) - 1)c_x$, so (9) is $\min\{\sum_x c_x \delta_x : \sum_x b_x \delta_x \geq \bar{b}, \delta \in [0, 1]^n\}$ with $\bar{b} := c - \Delta^{FI}$.

Step 1 (Support). If δ is feasible and $\delta_x > 0$ for some x with $b_x \leq 0$, replacing δ_x by 0 weakly increases $\sum_x b_x \delta_x$ and strictly decreases $\sum_x c_x \delta_x$. Hence any solution has $\delta_x = 0$ whenever $b_x \leq 0$, i.e. whenever $\ell(x) \leq 1$, and feasibility of (9) is equivalent to (10).

Step 2 (The constraint binds). If $\sum_x b_x \delta_x > \bar{b}$ at a feasible δ with $\delta \neq 0$, pick x with $\delta_x > 0$ and reduce δ_x slightly: feasibility is preserved and the objective strictly falls. Hence at any solution $\sum_x b_x \delta_x = \bar{b}$ (the case $\bar{b} \leq 0$ is degenerate and gives $\delta^* = 0$).

Step 3 (Exchange). Suppose δ is feasible, $\ell(x) > \ell(y) > 1$, $\delta_x < 1$ and $\delta_y > 0$. For $\theta > 0$ small set $\delta'_x = \delta_x + \theta/b_x$, $\delta'_y = \delta_y - \theta/b_y$, and $\delta' = \delta$ elsewhere; then $\delta' \in [0, 1]^n$ and $\sum_z b_z \delta'_z = \sum_z b_z \delta_z$, so δ' is feasible. The change in the objective is

$$\theta \left(\frac{c_x}{b_x} - \frac{c_y}{b_y} \right) = \theta \left(\frac{1}{\ell(x) - 1} - \frac{1}{\ell(y) - 1} \right) < 0,$$

because $t \mapsto 1/(t - 1)$ is strictly decreasing on $(1, \infty)$. Hence δ is not optimal. Therefore at any solution, $\ell(x) > \ell(y) > 1$ and $\delta_y > 0$ imply $\delta_x = 1$.

Step 4 (Conclusion). Steps 1–3 show that any solution fills the states with $\ell > 1$ in weakly descending order of ℓ until the constraint holds with equality. This is exactly δ^* , and δ^* is feasible by (10). Among states with distinct values of ℓ , at most the marginal state y_j carries $\delta_{y_j}^* \in (0, 1)$. When several states share the value $\ell(y_j)$ the solution set contains profiles with several fractional coordinates, but δ^* as constructed has at most one; if ℓ is injective on $\{\ell > 1\}$, Step 3 forces the fill order and Step 2 the stopping point, so δ^* is unique. $\square$

Remark 2 (Scope). *Program (9) is a relaxation of (8): it imposes on-path obedience implicitly through α^{FI} but drops the requirements $Q_1^W \geq 0$, $Q_1^S \leq 0$, $Q_0^W \geq 0$, $Q_0^S \leq 0$ that define the regular region, and it uses $V_0 = Q_0^W$ rather than $V_0 = [Q_0^W]^+ + [Q_0^S]^+$. A solution of (9) therefore solves (8) only when the induced α satisfies those four inequalities, which must be checked. Lemma 3 below verifies them in the environment of Corollary 1, and Proposition 8 verifies them in closed form in the three-state example.*

The magnitude $|q(x)|$ cancels from the ranking. What matters is how diagnostic the state is of past shirking. A state with high $\ell(x)$ is common after shirking but rare after effort. Pooling it into a continuation message has little effect on the worker's information but strongly contaminates the same message for the shirker.

7.1 Disconnected pooling in the spatial model

Order the states $x_1 < x_2 < \dots < x_n$ and abbreviate $f_i(x_k)$ by $f_i(k)$, $q(x_k)$ by $q(k)$ and $\ell(x_k)$ by $\ell(k)$.

Assumption 3 (Spatial environment). (i) (*Interior continuation region.*) $q(k) \neq 0$ for all k , and there are indices $L \leq H$ with $2 \leq L$ and $H \leq n$ such that $q(k) > 0$ if and only if $L \leq k \leq H$.

(ii) (*Strict monotone likelihood ratio.*) $\ell(1) > \ell(2) > \dots > \ell(n)$, equivalently f_1/f_0 is strictly increasing.

(iii) (*Diagnostic lower tail.*) $\ell(L-1) > 1$.

Write $N^- := \{1, \dots, L-1\}$ for the lower tail, $b_k := [f_0(k) - f_1(k)]|q(k)|$ for the incentive benefit of distorting state k , and $\Lambda_m := \sum_{k \leq m} b_k$ for $m \leq L-1$.

By (ii) and (iii), $\ell(k) > 1$ and hence $b_k > 0$ for every $k \in N^-$, and $\ell(k) < \ell(L-1)$ for every $k \geq L$: every lower-tail state is strictly more shirking-diagnostic than every state in or above the continuation region.

For $s \in [0, \Lambda_{L-1}]$ let $\delta(s)$ denote the greedy profile of Proposition 4 with budget $\bar{b} = s$ restricted to N^- , and let $\alpha(s)$ be the associated experiment, so that

$$\alpha_k(s) = \delta_k(s) \ (k \in N^-), \quad \alpha_k(s) = 1 \ (L \leq k \leq H), \quad \alpha_k(s) = 0 \ (k > H). \quad (11)$$

Lemma 3 (Validity of the regular region on the lower-tail path). *Under Assumption 3, along the path (11):*

- (i) $Q_1^S(\alpha(s)) \leq 0$ and $Q_0^S(\alpha(s)) \leq 0$ for every $s \in [0, \Lambda_{L-1}]$;
- (ii) $s \mapsto Q_0^W(\alpha(s))$ is continuous, piecewise linear and strictly decreasing, with $Q_0^W(\alpha(0)) = \sum_{k=L}^H f_0(k)q(k) > 0$;
- (iii) $Q_1^W(\alpha(s)) - Q_0^W(\alpha(s)) = \Delta^{FI} + s$, so $Q_0^W(\alpha(s)) \geq 0$ and $s > -\Delta^{FI}$ imply $Q_1^W(\alpha(s)) > 0$.

Consequently, defining

$$\bar{s} := \max \{s \in [0, \Lambda_{L-1}] : Q_0^W(\alpha(s)) \geq 0\} > 0, \quad \bar{c} := \Delta^{FI} + \bar{s},$$

the profile $\alpha(s)$ lies in the regular region for every $s \in [0, \bar{s}]$; and for $s \in (\bar{s}, \Lambda_{L-1}]$ the attained incentive wedge $Q_1^W(\alpha(s)) - [Q_0^W(\alpha(s))]^+ = Q_1^W(\alpha(s))$ is strictly decreasing in s , so no profile of the form (11) delivers a wedge exceeding $\bar{c}$.

Proof. (i) By (11), $1 - \alpha_k(s) = 0$ for $L \leq k \leq H$, so for $i \in \{0, 1\}$

$$Q_i^S(\alpha(s)) = \sum_{k \notin \{L, \dots, H\}} f_i(k) [1 - \alpha_k(s)] q(k) \leq 0,$$

since every term has $q(k) < 0$ by Assumption 3(i) and $1 - \alpha_k(s) \geq 0$.

(ii) By (11),

$$Q_0^W(\alpha(s)) = \sum_{k=L}^H f_0(k) q(k) - \sum_{k \in N^-} f_0(k) |q(k)| \delta_k(s),$$

which is positive at $s = 0$ because $q(k) > 0$ on $\{L, \dots, H\}$ and $f_0 > 0$. Along the greedy path, increasing s by ds raises δ_k by ds/b_k for the single currently marginal state k , so Q_0^W falls at rate $f_0(k) |q(k)| / b_k > 0$; the rate is piecewise constant in s , changing only at the finitely many indices where the marginal state changes.

(iii) By the two displays preceding (9), $Q_1^W - Q_0^W = \Delta^{FI} + \sum_k b_k \delta_k$, and along (11) the sum equals s by construction.

The set defining $\bar{s}$ is closed by continuity and contains 0 by (ii), so the maximum is attained and $\bar{s} > 0$. For $s \leq \bar{s}$, part (ii) gives $Q_0^W(\alpha(s)) \geq 0$ and part (iii) gives $Q_1^W(\alpha(s)) = Q_0^W(\alpha(s)) + \Delta^{FI} + s$; together with (i) this places $\alpha(s)$ in the regular region for every s with $\Delta^{FI} + s \geq 0$. In the application below $s = c - \Delta^{FI}$ with $c \geq 0$, so $\Delta^{FI} + s = c \geq 0$ and the qualification is automatic. For $s > \bar{s}$, $Q_0^W < 0$, so the shirker chooses $b = 0$ after both messages by (i), $V_0 = 0$, and the wedge equals $Q_1^W(\alpha(s)) = \sum_{k=L}^H f_1(k) q(k) - \sum_{k \in N^-} f_1(k) |q(k)| \delta_k(s)$, strictly decreasing in s . Its value at $s = \bar{s}$ is $\Delta^{FI} + \bar{s} = \bar{c}$ by (iii) and $Q_0^W(\alpha(\bar{s})) = 0$. $\square$

Corollary 1 (Lower-tail contamination and disconnected pooling). *Let Assumption 3 hold and suppose the effort cost satisfies*

$$\Delta^{FI} < c \leq \bar{c}. \quad (12)$$

Let $j := \min\{m \leq L - 1 : \Lambda_m \geq c - \Delta^{FI}\}$. Then the experiment $\alpha^ := \alpha(c - \Delta^{FI})$ solves (8); explicitly,*

$$\alpha_k^* = 1 \quad (k < j), \quad \alpha_j^* = \frac{c - \Delta^{FI} - \Lambda_{j-1}}{b_j} \in (0, 1], \quad \alpha_k^* = 0 \quad (j < k < L),$$

$$\alpha_k^* = 1 \quad (L \leq k \leq H), \quad \alpha_k^* = 0 \quad (k > H),$$

so the set of states sent to the work message with positive probability is

$$W = \{x_1, \dots, x_j\} \cup \{x_L, \dots, x_H\}.$$

This set is disconnected if and only if $j \leq L - 2$, which holds if and only if $L \geq 3$ and

$$c \leq \Delta^{FI} + \Lambda_{L-2}. \quad (13)$$

No state above x_H and no state in the continuation region is ever distorted.

Proof. Step 1 (The greedy profile is supported on the lower tail). By Proposition 4, a solution of (9) distorts states with $\ell > 1$ in descending order of ℓ until the budget $c - \Delta^{FI} > 0$ is exhausted. By Assumption 3(ii)–(iii), $\min_{k \in N^-} \ell(k) = \ell(L - 1) > \ell(k)$ for every $k \geq L$, so all states of N^- precede every other state in that order. By Lemma 3, $c \leq \bar{c}$ gives $c - \Delta^{FI} \leq \bar{s} \leq \Lambda_{L-1}$, so the budget is exhausted within N^- and $\delta_k^* = 0$ for every $k \geq L$. The index j and the stated values

of α^* are then read off from Proposition 4 together with the definition $\alpha_x = 1 - \delta_x$ on $\{q > 0\}$ and $\alpha_x = \delta_x$ on $\{q < 0\}$.

Step 2 (α^* is feasible in (8)). $\alpha^* = \alpha(c - \Delta^{FI})$ with $c - \Delta^{FI} \leq \bar{s}$, so by Lemma 3 it lies in the regular region: $Q_1^W \geq 0$, $Q_1^S \leq 0$, $Q_0^W \geq 0$, $Q_0^S \leq 0$. Hence $V_1 = Q_1^W$ and $V_0 = Q_0^W$, and the incentive wedge equals $\Delta^{FI} + (c - \Delta^{FI}) = c$ by Lemma 3(iii). Taking $z_W = Q_0^W$ and $z_S = 0$, the profile is feasible in (8).

Step 3 (Optimality). Consider any α feasible in (8) and let δ be the associated distortion profile. If $Q_0^W(\alpha) \geq 0$ and $Q_0^S(\alpha) \leq 0$, then by Lemma 2 the incentive constraint reads $\Delta^{FI} + \sum_k b_k \delta_k \geq c$, so δ is feasible in (9) and, by Proposition 4,

$$\sum_k f_1(k) |q(k)| \delta_k \geq \sum_k f_1(k) |q(k)| \delta_k^*,$$

whence $Q_1^W(\alpha) = V_1^{FI} - \sum_k f_1(k) |q(k)| \delta_k \leq Q_1^W(\alpha^*)$.

Otherwise $[Q_0^W(\alpha)]^+ + [Q_0^S(\alpha)]^+ > Q_0^W(\alpha)$, and the incentive constraint of (8) implies $Q_1^W(\alpha) - Q_0^W(\alpha) > c$, i.e. $\Delta^{FI} + \sum_k b_k \delta_k > c$. Then δ is again feasible in (9) and the same inequality gives $Q_1^W(\alpha) \leq Q_1^W(\alpha^*)$.

In both cases the objective at α is weakly below its value at α^* , so α^* is optimal.

Step 4 (Disconnectedness). By Step 1, $\alpha_k^* > 0$ exactly on $\{1, \dots, j\} \cup \{L, \dots, H\}$. This set is an interval of $\{1, \dots, n\}$ if and only if $j \geq L - 1$; since $j \leq L - 1$ by construction, it is disconnected if and only if $j \leq L - 2$, which requires $L \geq 3$ and, by the definition of j , is equivalent to $\Lambda_{L-2} \geq c - \Delta^{FI}$. $\square$

The lower-tail states are not pooled in order to persuade a common receiver that continuation is productive; adding them strictly lowers the expected continuation payoff of both types. They are pooled because, by Assumption 3(ii)–(iii), they are far more likely after shirking than after effort, so the same work message is contaminated much more for the shirker than for the worker.

7.2 The global implementability frontier

Corollary 1 characterises the optimum for $c \leq \bar{c}$, where $\bar{c}$ was defined as the largest wedge attainable along the lower-tail path (11). One further condition makes $\bar{c}$ the global frontier.

Assumption 4 (Sharp lower tail). $\ell(L) < 1$.

Under Assumptions 3 and 4, ℓ is strictly decreasing and crosses 1 strictly between x_{L-1} and x_L , so

$$\{k : \ell(k) > 1\} = N^-, \quad (14)$$

and $\Delta^{FI} = \sum_{k=L}^H f_1(k) [1 - \ell(k)] q(k) > 0$: shirking is diagnosed by exactly the states below the continuation region, and the efficient signal already delivers a strictly positive wedge.

The key observation is that the two knapsack problems appearing in the analysis share an order.

Lemma 4 (Common greedy order). Let $c_k := f_1(k) |q(k)|$, $b_k := [f_0(k) - f_1(k)] |q(k)|$ and $w_k := f_0(k) |q(k)|$, all evaluated at states with $q(k) \neq 0$. Then

$$\frac{b_k}{c_k} = \ell(k) - 1, \quad \frac{b_k}{w_k} = 1 - \frac{1}{\ell(k)},$$

and both are strictly increasing functions of $\ell(k)$ on $(0, \infty)$. Hence the minimum-cost problem (9) and the maximum-benefit problem $\max\{\sum_k b_k \delta_k : \sum_k w_k \delta_k \leq \kappa, \delta \in [0, 1]^n\}$ are both solved by filling states in strictly descending order of ℓ .

Proof. The two identities are immediate from $\ell(k) = f_0(k)/f_1(k)$ and $f_1(k)|q(k)| > 0$. The map $t \mapsto t - 1$ is strictly increasing and $t \mapsto 1 - 1/t$ is strictly increasing on $(0, \infty)$. In a fractional knapsack the greedy order is by benefit per unit of the constrained resource, which is b_k/c_k in the first problem and b_k/w_k in the second; both rank states by $\ell(k)$. $\square$

Proposition 5 (Implementability frontier). *Under Assumptions 3 and 4,*

$$\max_{\alpha \in [0,1]^n} \{Q_1^W - [Q_0^W]^+ - [Q_0^S]^+ : Q_1^W \geq 0, Q_1^S \leq 0\} = \bar{c}.$$

Hence $a^* = 1$ is implementable if and only if $c \leq \bar{c}$, and Corollary 1 characterises the optimal experiment on the entire implementable range $c \in (\Delta^{FI}, \bar{c}]$.

Proof. Write $\kappa := \sum_{k=L}^H f_0(k)q(k) > 0$, so that $Q_0^W = \kappa - \sum_k w_k \delta_k$ and, by the displays preceding (9), $Q_1^W - Q_0^W = \Delta^{FI} + \sum_k b_k \delta_k$. Define

$$\bar{c}^A := \max \left\{ \Delta^{FI} + \sum_k b_k \delta_k : \sum_k w_k \delta_k \leq \kappa, \delta \in [0,1]^n \right\}.$$

Step 1 ($\bar{c}^A$ is an upper bound in every regime). If $Q_0^W \geq 0$ and $Q_0^S \leq 0$, the wedge equals $Q_1^W - Q_0^W$ and $Q_0^W \geq 0$ is exactly $\sum_k w_k \delta_k \leq \kappa$, so the wedge is at most $\bar{c}^A$. If $Q_0^W \geq 0$ and $Q_0^S > 0$, the wedge equals $Q_1^W - Q_0^W - Q_0^S < Q_1^W - Q_0^W \leq \bar{c}^A$. If $Q_0^W < 0$, the wedge is at most Q_1^W ; consider $\max\{Q_1^W : \sum_k w_k \delta_k \geq \kappa, \delta \in [0,1]^n\}$. Since $Q_1^W = V_1^{FI} - \sum_k c_k \delta_k$ is strictly decreasing in each coordinate and $w_k > 0$, at any maximiser the resource constraint holds with equality: otherwise some $\delta_k > 0$ (such k exists because $\kappa > 0$) could be reduced, raising Q_1^W while preserving feasibility. At such a point $Q_0^W = 0$ and the wedge is $Q_1^W - 0 = Q_1^W - Q_0^W \leq \bar{c}^A$.

Step 2 ($\bar{c}^A = \bar{c}$). By Lemma 4 the maximisation defining $\bar{c}^A$ is solved by filling states in strictly descending order of ℓ , using only states with $b_k > 0$, that is, with $\ell(k) > 1$. By (14) those are exactly the states of N^- , and descending ℓ is ascending k by Assumption 3(ii). The fill therefore traces the path $\alpha(s)$ of (11) and stops either when the resource constraint binds, that is when $Q_0^W = 0$, or when $\delta_k = 1$ for every $k \in N^-$. In the first case the accumulated benefit is $\bar{s}$ by definition of $\bar{s}$; in the second it is Λ_{L-1} , and then $Q_0^W(\alpha(\Lambda_{L-1})) \geq 0$ so $\bar{s} = \Lambda_{L-1}$ as well. Either way $\bar{c}^A = \Delta^{FI} + \bar{s} = \bar{c}$.

Step 3 (Attainment). The maximiser constructed in Step 2 is $\alpha(\bar{s})$, whose distortion is supported on N^- . By Lemma 3(i), $Q_1^S \leq 0$ and $Q_0^S \leq 0$; by construction $Q_0^W \geq 0$; and $Q_1^W = Q_0^W + \bar{c} \geq \bar{c} > 0$ because $\Delta^{FI} > 0$. Hence $\alpha(\bar{s})$ satisfies the on-path obedience constraints and attains the wedge $\bar{c}$, which by Steps 1–2 is maximal. The implementability claim follows, and Corollary 1 applies on all of $(\Delta^{FI}, \bar{c}]$. $\square$

7.3 Comparative statics

Let

$$C(\beta) := \min \left\{ \sum_k c_k \delta_k : \sum_k b_k \delta_k \geq \beta, \delta \in [0,1]^n \right\}, \quad \beta \in [0, \Lambda_{L-1}],$$

denote the value of (9) as a function of the required incentive benefit, and let $\lambda(c)$ be the multiplier on the incentive constraint in (8).

Proposition 6 (Monotonicity in the effort cost). *Under Assumptions 3 and 4, on the implementable range $c \in (\Delta^{FI}, \bar{c}]$:*

- (i) the optimal distortion $\delta^*(c)$ is non-decreasing in c coordinatewise, and the cutoff $j(c)$ of Corollary 1 is non-decreasing;

- (ii) $V_1(c) = V_1^{FI} - C(c - \Delta^{FI})$ is continuous, strictly decreasing and concave in c ;
- (iii) $\lambda(c) = 1/[\ell(x_{j(c)}) - 1]$, which is non-decreasing in c and strictly increasing at each cutoff change;
- (iv) the work message is disconnected for $c \in (\Delta^{FI}, \Delta^{FI} + \Lambda_{L-2}]$ and connected for larger c : raising the effort cost fills in the gap between the contaminated lower tail and the continuation region.

Proof. (i) By Proposition 4 the solution is the greedy fill to benefit level $\beta = c - \Delta^{FI}$, and the greedy profile is coordinatewise non-decreasing in β : raising β leaves $\delta_{y_i}^* = 1$ for $i < j$ unchanged, weakly raises $\delta_{y_j}^*$, and can only add further states. The cutoff j is the least index with $\Lambda_j \geq \beta$, non-decreasing in β .

(ii) C is the value of a minimisation linear program parameterised by the right-hand side β , hence convex and non-decreasing in β ; it is finite and continuous on $[0, \Lambda_{L-1}]$ by Proposition 4, and strictly increasing because $c_k > 0$. By Corollary 1, $V_1(c) = Q_1^W(\alpha^*) = V_1^{FI} - C(c - \Delta^{FI})$, which is therefore continuous, strictly decreasing and concave in c .

(iii) $\lambda(c)$ is the right derivative of the optimal-cost function, $\lambda(c) = C'(c - \Delta^{FI})$. Along the greedy the marginal state at benefit level β is y_j , and one extra unit of benefit costs $c_{y_j}/b_{y_j} = 1/[\ell(y_j) - 1]$ by Lemma 4. Since $\ell(y_j)$ is non-increasing in j and j is non-decreasing in c by (i), λ is non-decreasing; it jumps upward exactly when j increases. Monotonicity also follows from convexity of C in (ii).

(iv) Immediate from criterion (13) of Corollary 1 and monotonicity of j in (i). $\square$

The economics of (ii)–(iii) is that the principal spends her cheapest contamination first. Each additional unit of incentive must be bought with a state that is less diagnostic of shirking, so the marginal informational price of incentives rises and the on-path value is concave in the cost being deterred.

Proposition 7 (Monotonicity in detectability). *Hold f_1 and q fixed and replace f_0 by $\tilde{f}_0$, with $\tilde{\ell} := \tilde{f}_0/f_1$, such that Assumptions 3 and 4 continue to hold and*

$$\tilde{\ell}(k) \geq \ell(k) \quad (k \in N^-), \quad \tilde{\ell}(k) \leq \ell(k) \quad (L \leq k \leq H).$$

Then $\tilde{\Delta}^{FI} \geq \Delta^{FI}$, $\tilde{C}(\beta) \leq C(\beta)$ for every β , and therefore $\tilde{V}_1(c) \geq V_1(c)$ for every c implementable under both primitives.

Proof. Since $q > 0$ on $\{L, \dots, H\}$ and $\tilde{\ell} \leq \ell$ there, $\tilde{\Delta}^{FI} = \sum_{k=L}^H f_1(k)[1 - \tilde{\ell}(k)]q(k) \geq \Delta^{FI}$. The costs $c_k = f_1(k)|q(k)|$ are unchanged, while $\tilde{b}_k = f_1(k)[\tilde{\ell}(k) - 1]|q(k)| \geq b_k$ for $k \in N^-$; by Assumption 4 and (14) the greedy uses only N^- under both primitives. Hence every δ feasible for the constraint $\sum_k b_k \delta_k \geq \beta$ is feasible for $\sum_k \tilde{b}_k \delta_k \geq \beta$ at the same cost, so $\tilde{C}(\beta) \leq C(\beta)$. Finally $V_1^{FI} = \sum_{k=L}^H f_1(k)q(k)$ does not depend on f_0 , and C is non-decreasing, so

$$\tilde{V}_1(c) = V_1^{FI} - \tilde{C}(c - \tilde{\Delta}^{FI}) \geq V_1^{FI} - C(c - \Delta^{FI}) = V_1(c). \quad \square$$

8 A three-state closed-form example

Let $X = \{1, 2, 3\}$ and define

$$f_1 = (\varepsilon, p, \eta), \quad f_0 = (\eta, p, \varepsilon), \quad \eta := 1 - p - \varepsilon,$$

with $p \in (0, 1)$ and $0 < \varepsilon < \eta$. (The constant is written η rather than A to avoid a collision with the initial-action set A of Section 2.) Effort shifts probability mass from state 1 to state 3 while leaving the probability of state 2 unchanged.

Let

$$q(1) = -L, \quad q(2) = G, \quad q(3) = -H,$$

where $L, G, H > 0$. Only state 2 warrants continuation. Consider the signal

$$\mathbb{P}(W \mid 2) = 1, \quad \mathbb{P}(W \mid 3) = 0, \quad \mathbb{P}(W \mid 1) = t,$$

which in the notation of Corollary 1 is $\alpha = (t, 1, 0)$: the principal sometimes pools the hopeless state 1 with the productive state 2. Define

$$\bar{c} := (\eta - \varepsilon)L \min \left\{ 1, \frac{pG}{\eta L} \right\}.$$

Proposition 8 (Optimal distortion in the three-state example). *The target action $a^* = 1$ is implementable if and only if $0 \leq c \leq \bar{c}$. For every such c , the unique solution of (8) is $\alpha^* = (t^*, 1, 0)$ with*

$$t^* = \frac{c}{(\eta - \varepsilon)L},$$

the induced continuation values are $V_1 = pG - \varepsilon Lt^*$ and $V_0 = pG - \eta Lt^*$, and the on-path continuation loss relative to the efficient signal is

$$V_1^{FI} - V_1(t^*) = \frac{\varepsilon c}{\eta - \varepsilon}.$$

Proof. For $\alpha \in [0, 1]^3$,

$$Q_1^W = -\varepsilon L \alpha_1 + pG \alpha_2 - \eta H \alpha_3, \quad Q_0^W = -\eta L \alpha_1 + pG \alpha_2 - \varepsilon H \alpha_3, \quad (15)$$

$$Q_1^S = -\varepsilon L(1 - \alpha_1) + pG(1 - \alpha_2) - \eta H(1 - \alpha_3), \quad Q_0^S = -\eta L(1 - \alpha_1) + pG(1 - \alpha_2) - \varepsilon H(1 - \alpha_3). \quad (16)$$

Subtracting,

$$Q_1^W - Q_0^W = (\eta - \varepsilon)L \alpha_1 - (\eta - \varepsilon)H \alpha_3. \quad (17)$$

The proof has four steps: the candidate is feasible; it is optimal; no $c > \bar{c}$ is implementable; uniqueness.

Step 1 (The candidate). Here $\ell = (\eta/\varepsilon, 1, \varepsilon/\eta)$ is strictly decreasing because $\varepsilon < \eta$, $q > 0$ exactly on $\{2\}$, and $\ell(1) = \eta/\varepsilon > 1$, so Assumption 3 holds with $L = H = 2$, $N^- = \{1\}$. Moreover

$$\Delta^{FI} = [f_1(2) - f_0(2)] q(2) = (p - p)G = 0, \quad b_1 = (\eta - \varepsilon)L, \quad b_2 = 0, \quad b_3 = (\varepsilon - \eta)H < 0,$$

so state 1 is the only state with strictly positive incentive benefit. Proposition 4 therefore gives $\delta^* = (\min\{1, c/b_1\}, 0, 0)$, i.e. $\alpha^* = (t^*, 1, 0)$ with $t^* = c/[(\eta - \varepsilon)L]$, feasible in (9) if and only if $c \leq (\eta - \varepsilon)L$.

At α^* , (15)–(16) give

$$Q_1^W = pG - \varepsilon Lt^*, \quad Q_0^W = pG - \eta Lt^*, \quad Q_1^S = -\varepsilon L(1 - t^*) - \eta H < 0, \quad Q_0^S = -\eta L(1 - t^*) - \varepsilon H < 0.$$

Hence $Q_0^W \geq 0$ if and only if $t^* \leq pG/(\eta L)$, and in that case $Q_1^W > Q_0^W \geq 0$ because $\varepsilon < \eta$ and $t^* > 0$ (for $c > 0$; for $c = 0$ the claim is immediate). Combining the two restrictions $t^* \leq 1$ and $t^* \leq pG/(\eta L)$ gives exactly $c \leq \bar{c}$, and on that range α^* lies in the regular region with $V_1 - V_0 = Q_1^W - Q_0^W = (\eta - \varepsilon)Lt^* = c$, so α^* is feasible in (8).

Step 2 (Optimality). Let α be any profile feasible in (8); by Lemma 2 its wedge $Q_1^W - [Q_0^W]^+ - [Q_0^S]^+$ is at least c . Three cases exhaust the possibilities.

Case A: $Q_0^W \geq 0$ and $Q_0^S \leq 0$. The wedge is $Q_1^W - Q_0^W$, so (17) gives $L\alpha_1 - H\alpha_3 \geq c/(\eta - \varepsilon) = Lt^*$, i.e. $\alpha_1 \geq t^* + (H/L)\alpha_3$. Substituting into (15) and using $\alpha_2 \leq 1$,

$$Q_1^W \leq pG - \varepsilon L \left(t^* + \frac{H}{L} \alpha_3 \right) - \eta H \alpha_3 = pG - \varepsilon L t^* - (\varepsilon + \eta) H \alpha_3 \leq pG - \varepsilon L t^*.$$

Case B: $Q_0^W \geq 0$ and $Q_0^S > 0$. From (16), $Q_0^S > 0$ forces $pG(1 - \alpha_2) > \eta L(1 - \alpha_1) + \varepsilon H(1 - \alpha_3) \geq \eta L(1 - \alpha_1)$, hence $pG\alpha_2 < pG - \eta L(1 - \alpha_1)$. Substituting into (15),

$$Q_1^W < pG - \eta L(1 - \alpha_1) - \varepsilon L \alpha_1 - \eta H \alpha_3 \leq pG - \eta L + (\eta - \varepsilon) L \alpha_1 \leq pG - \varepsilon L \leq pG - \varepsilon L t^*,$$

where the last two inequalities use $\alpha_1 \leq 1$ and $t^* \leq 1$.

Case C: $Q_0^W < 0$. Then $\eta L \alpha_1 > pG\alpha_2 - \varepsilon H \alpha_3$, so $\varepsilon L \alpha_1 > (\varepsilon/\eta)(pG\alpha_2 - \varepsilon H \alpha_3)$ and, by (15),

$$Q_1^W < pG\alpha_2 - \eta H \alpha_3 - \frac{\varepsilon}{\eta} (pG\alpha_2 - \varepsilon H \alpha_3) = \frac{\eta - \varepsilon}{\eta} pG\alpha_2 - \frac{\eta^2 - \varepsilon^2}{\eta} H \alpha_3 \leq \frac{\eta - \varepsilon}{\eta} pG.$$

Since $c \leq \bar{c}$ implies $t^* \leq pG/(\eta L)$, we have $\varepsilon L t^* \leq \varepsilon pG/\eta$ and therefore $pG - \varepsilon L t^* \geq pG(\eta - \varepsilon)/\eta > Q_1^W$.

In every case $Q_1^W(\alpha) \leq pG - \varepsilon L t^* = Q_1^W(\alpha^*)$, so α^* solves (8).

Step 3 (Implementability frontier). Let α satisfy on-path obedience. In Case A the wedge is $(\eta - \varepsilon)(L\alpha_1 - H\alpha_3) \leq (\eta - \varepsilon)L\alpha_1$, and $Q_0^W \geq 0$ with $\alpha_2 \leq 1$, $\alpha_3 \geq 0$ forces $\eta L \alpha_1 \leq pG$; hence the wedge is at most $(\eta - \varepsilon)L \min\{1, pG/(\eta L)\} = \bar{c}$. In Case B the constraint $Q_0^W \geq 0$ still holds, so the same argument gives $\eta L \alpha_1 \leq pG$ and hence $Q_1^W - Q_0^W \leq \bar{c}$; since $Q_0^S > 0$, the wedge equals $Q_1^W - Q_0^W - Q_0^S < Q_1^W - Q_0^W \leq \bar{c}$. In Case C the wedge is at most Q_1^W , and the two bounds derived in Case C of Step 2 give $Q_1^W < (\eta - \varepsilon)L$ (from (17) with $Q_0^W < 0$) and $Q_1^W < (\eta - \varepsilon)pG/\eta$, hence $Q_1^W < \bar{c}$. The bound $\bar{c}$ is attained in Case A at $\alpha = (\min\{1, pG/(\eta L)\}, 1, 0)$.

Step 4 (Uniqueness and the loss formula). Case A of Step 2 shows $Q_1^W = Q_1^W(\alpha^*)$ requires $\alpha_3 = 0$, $\alpha_2 = 1$ and $\alpha_1 = t^*$. Cases B and C give strict inequalities. Finally $V_1^{FI} = pG$, obtained at $t = 0$, so

$$V_1^{FI} - V_1(t^*) = \varepsilon L t^* = \frac{\varepsilon L c}{(\eta - \varepsilon)L} = \frac{\varepsilon c}{\eta - \varepsilon}. \quad \square$$

The informational cost of incentives is small when state 1 is rare after effort but common after shirking. In the limiting case $\varepsilon \rightarrow 0$, the partition becomes

$$W = \{1, 2\}, \quad S = \{3\}.$$

For the worker, W identifies $x = 2$ and S identifies $x = 3$, so the signal is fully informative. For the shirker, W leaves $x \in \{1, 2\}$ unresolved. This is the perfect-complementarity benchmark.

Remark 3 (Why the pool is connected here). *With $n = 3$ and $L = H = 2$ there is only one state below the continuation region, so $j = L - 1 = 1$ and the criterion (13) of Corollary 1 fails: disconnected pooling requires at least two states strictly below x_L . The three-state example is the smallest environment exhibiting lower-tail contamination, and the smallest exhibiting disconnected lower-tail contamination has $n = 4$ with $L = 3$.*

8.1 The principal's implementation decision

Suppose the direct state payoff is

$$R(x) = \beta x, \quad \beta > 0.$$

Proposition 9 (When to induce effort). *Let $R(x) = \beta x$ with $\beta > 0$ and suppose $0 \leq c \leq \bar{c}$, so that effort is implementable. Then the principal's payoff from optimally implementing $a^* = 1$ exceeds her payoff from implementing $a^* = 0$ if and only if*

$$2\beta(\eta - \varepsilon)^2 \geq \varepsilon c.$$

Proof. Step 1 (The shirking benchmark). Implementing $a^* = 0$ imposes the incentive constraint $V_0 \geq V_1 - c$, i.e. an upper bound on the wedge. The efficient signal $\alpha^{FI} = (0, 1, 0)$ delivers wedge $\Delta^{FI} = 0 \leq c$, so it is feasible for the target $a^* = 0$ and it is optimal for that target. It is optimal because, with $w(b, \cdot) = u(b, \cdot)$, the principal's continuation payoff under shirking equals the shirker's value $V_0 = [Q_0^W]^+ + [Q_0^S]^+$, and for every α

$$[Q_0^W]^+ + [Q_0^S]^+ \leq \sum_x f_0(x) \alpha_x [q(x)]^+ + \sum_x f_0(x) (1 - \alpha_x) [q(x)]^+ = \sum_x f_0(x) [q(x)]^+ = pG,$$

with equality at α^{FI} . Adding the direct term, the value of implementing $a^* = 0$ is $\mathbb{E}_{f_0}[\beta x] + pG$.

Step 2 (The effort value). By Proposition 8, implementing $a^* = 1$ optimally yields continuation payoff $Q_1^W(\alpha^*) = pG - \varepsilon c / (\eta - \varepsilon)$, since $w(b, \cdot) = u(b, \cdot)$ makes the principal's continuation payoff equal to V_1 . The value of implementing $a^* = 1$ is therefore $\mathbb{E}_{f_1}[\beta x] + pG - \varepsilon c / (\eta - \varepsilon)$. The agent's cost c does not enter the principal's payoff, as there are no transfers.

Step 3 (Comparison). With $f_1 = (\varepsilon, p, \eta)$ and $f_0 = (\eta, p, \varepsilon)$,

$$\mathbb{E}_{f_1}[x] - \mathbb{E}_{f_0}[x] = (\varepsilon + 2p + 3\eta) - (\eta + 2p + 3\varepsilon) = 2(\eta - \varepsilon).$$

Subtracting the Step-1 value from the Step-2 value gives

$$2\beta(\eta - \varepsilon) - \frac{\varepsilon c}{\eta - \varepsilon},$$

which is non-negative if and only if $2\beta(\eta - \varepsilon)^2 \geq \varepsilon c$, using $\eta > \varepsilon$. □

9 Which deviations bind

Return to a general finite action set $A = \{0, 1, \dots, K-1\}$, ordered so that $k(0) < k(1) < \dots < k(K-1)$, with target $a^* = K-1$ and binary continuation as in Section 6. Agenda item 2 asks for conditions under which only the *adjacent* deviation $a = a^* - 1$ binds. This section shows that a mild condition delivers the sharper conclusion that exactly *one* deviation binds, but that the binding deviation need not be the adjacent one, even under monotone likelihood ratios and strictly convex costs.

Remark 4 (Two messages still suffice). *The proof of Proposition 3 uses only that merging preserves the target type's expected continuation payoff and, by subadditivity of $z \mapsto [z]^+$, weakly lowers the optimised value of every other type. It therefore applies verbatim for any finite A : for each $a \neq a^*$, $V_a(\hat{\sigma}) \leq V_a(\sigma)$, so all $K-1$ incentive constraints are weakly relaxed simultaneously.*

Write $\ell_a := f_a/f_{a^*}$, $d_a := k(a^*) - k(a) > 0$, and, for a two-message experiment with distortion profile δ ,

$$c_k := f_{a^*}(k)|q(k)|, \quad b_{a,k} := [f_a(k) - f_{a^*}(k)]|q(k)| = c_k[\ell_a(k) - 1], \quad \Delta_a^{FI} := \sum_{q(k)>0} [f_{a^*}(k) - f_a(k)]q(k).$$

In the regular region, where every type obeys the recommendation, the incentive constraint against a is

$$\Delta_a^{FI} + \sum_k b_{a,k}\delta_k \geq d_a, \quad (18)$$

and the principal minimises $\sum_k c_k\delta_k$ subject to (18) for every $a \neq a^*$.

Assumption 5 (Ordered family with a common diagnostic set). States are ordered $x_1 < \dots < x_n$, $q(k) \neq 0$ for all k , and there is $L \geq 2$ such that:

- (i) ℓ_a is strictly decreasing in k for every $a \neq a^*$;
- (ii) $\ell_a(k) > 1$ for $k < L$ and $\ell_a(k) < 1$ for $k \geq L$, for every $a \neq a^*$.

Part (i) is implied by the monotone likelihood ratio property across the action set. Part (ii) says every deviation is diagnosed by the same lower tail $N^- = \{x_1, \dots, x_{L-1}\}$; it is the many-action version of Assumptions 3(iii) and 4.

For $t \in [0, L-1]$ let $\delta^{(t)}$ be the profile that fills N^- from the bottom to depth t , that is $\delta_k^{(t)} = 1$ for $k \leq \lfloor t \rfloor$, $\delta_{\lfloor t \rfloor + 1}^{(t)} = t - \lfloor t \rfloor$ and $\delta_k^{(t)} = 0$ otherwise, and set

$$\Gamma_a(t) := \sum_k b_{a,k}\delta_k^{(t)}, \quad t_a := \begin{cases} 0, & d_a \leq \Delta_a^{FI}, \\ \Gamma_a^{-1}(d_a - \Delta_a^{FI}), & \text{otherwise,} \end{cases}$$

which is well defined because Γ_a is continuous and strictly increasing on $[0, L-1]$ by Assumption 5(ii). Call t_a the *required depth* of deviation a .

Proposition 10 (Exactly one deviation binds). *Let Assumption 5 hold and suppose $t^\dagger := \max_{a \neq a^*} t_a \leq L-1$. Then, within the regular region, the optimal experiment is $\delta^{(t^\dagger)}$: the principal contaminates the lower tail from the bottom up to depth $t^\dagger$. The incentive constraint of deviation a binds if and only if $t_a = t^\dagger$; if the maximiser $a^\dagger$ is unique, exactly one constraint binds and all others are slack.*

Proof. Step 1 (A lower bound on cost). Relax the problem by keeping only constraint (18) for $a = a^\dagger$. By Assumption 5, $b_{a^\dagger,k} > 0$ exactly for $k \in N^-$, and $b_{a^\dagger,k}/c_k = \ell_{a^\dagger}(k) - 1$ is strictly decreasing in k by (i). Proposition 4 therefore gives as the unique solution of the relaxed problem the greedy fill of N^- from the bottom to the depth at which the accumulated benefit equals $d_{a^\dagger} - \Delta_{a^\dagger}^{FI}$, that is $\delta^{(t_a^\dagger)} = \delta^{(t^\dagger)}$. Its cost $\sum_k c_k\delta_k^{(t^\dagger)}$ is a lower bound for the full problem.

Step 2 ($\delta^{(t^\dagger)}$ is feasible). Fix $a \neq a^*$. By Assumption 5(ii), $b_{a,k} > 0$ for every $k \in N^-$, so Γ_a is strictly increasing on $[0, L-1]$. Since $t^\dagger \geq t_a$,

$$\Gamma_a(t^\dagger) \geq \Gamma_a(t_a) = \max\{d_a - \Delta_a^{FI}, 0\} \geq d_a - \Delta_a^{FI},$$

which is (18). Hence $\delta^{(t^\dagger)}$ satisfies every incentive constraint and attains the lower bound of Step 1, so it is optimal. The displayed inequality is an equality precisely when $t_a = t^\dagger$ (using strict monotonicity of Γ_a and $d_a > \Delta_a^{FI}$ at a binding constraint), which is the stated criterion. $\square$

Proposition 10 replaces the vector of multipliers in Proposition 2 by a single shadow price and reduces the many-action problem to the binary-action problem against the single deviation $a^\dagger$. What it does *not* deliver is that $a^\dagger = a^* - 1$.

Example 1 (The adjacent deviation need not bind). *Let $X = \{x_1, x_2, x_3\}$, $q = (-1, +1, -1)$, $A = \{0, 1, 2\}$ with $a^* = 2$, and*

$$f_2 = (0.10, 0.40, 0.50), \quad f_1 = (0.20, 0.40, 0.40), \quad f_0 = (0.25, 0.40, 0.35),$$

$$k(0) = 0, \quad k(1) = 0.04, \quad k(2) = 0.10.$$

The family is strictly ordered by the likelihood ratio, since $f_2/f_1 = (0.5, 1, 1.25)$ and $f_1/f_0 = (0.8, 1, 8/7)$ are strictly increasing, and the cost function is strictly convex, with increments 0.04 and 0.06. Moreover $\ell_1 = (2, 1, 0.8)$ and $\ell_0 = (2.5, 1, 0.7)$ are strictly decreasing with $\{\ell_a > 1\} = \{x_1\}$ for both deviations, so Assumption 5 holds with $L = 2$.

Because all three distributions put mass 0.40 on the only productive state, $\Delta_1^{FI} = \Delta_0^{FI} = 0$, and $N^- = \{x_1\}$ is a single lever, so $t_a = d_a/b_{a,1}$ with $b_{a,1} = f_a(x_1) - f_2(x_1)$:

$$t_1 = \frac{k(2) - k(1)}{f_1(x_1) - f_2(x_1)} = \frac{0.06}{0.10} = 0.60, \quad t_0 = \frac{k(2) - k(0)}{f_0(x_1) - f_2(x_1)} = \frac{0.10}{0.15} = \frac{2}{3}.$$

Hence $a^\dagger = 0$: the optimum is $\alpha^ = (2/3, 1, 0)$, the non-adjacent constraint binds exactly, and the adjacent constraint is slack by $0.10 \times \frac{2}{3} - 0.06 = 1/150$. Solving the linear program of Remark 4 directly returns $\alpha^* = (2/3, 1, 0)$ with principal value $1/3$, confirming both the optimum and the pattern of slacks.*

The mechanism behind Example 1 is transparent. Deterring a distant deviation requires a larger wedge, d_a being decreasing in a , but a distant deviation is also easier to detect, $b_{a,1}$ being decreasing in a . Which effect dominates is a race between two decreasing sequences, and convexity of k disciplines only the first. This yields the condition that agenda item 2 was asking for.

Corollary 2 (Single crossing for adjacency). *Let Assumption 5 hold with $L = 2$, so that $N^- = \{x_1\}$. The adjacent deviation binds, $a^\dagger = a^* - 1$, if and only if*

$$\frac{d_{a^*-1} - \Delta_{a^*-1}^{FI}}{f_{a^*-1}(x_1) - f_{a^*}(x_1)} \geq \frac{d_a - \Delta_a^{FI}}{f_a(x_1) - f_{a^*}(x_1)} \quad \text{for every } a < a^* - 1, \quad (19)$$

that is, if and only if the net incentive requirement $d_a - \Delta_a^{FI}$ falls across deviations at least as fast as detectability $f_a(x_1) - f_{a^}(x_1)$. Neither the monotone likelihood ratio property nor convexity of k implies (19).*

Proof. With $N^- = \{x_1\}$ the required depth is $t_a = [d_a - \Delta_a^{FI}]^+/b_{a,1}$ and $b_{a,1} = f_a(x_1) - f_{a^*}(x_1) > 0$ by Assumption 5(ii). By Proposition 10, $a^\dagger = a^* - 1$ if and only if $t_{a^*-1} \geq t_a$ for all a , which is (19). Example 1 satisfies the monotone likelihood ratio property and strict convexity of k while violating (19), since $0.60 < 2/3$. $\square$

Remark 5 (Beyond a single lever). *When $L > 2$ the required depths t_a are no longer ratios but inverses of the strictly increasing functions Γ_a , and (19) generalises to $t_{a^*-1} \geq t_a$ for all a . In 815 randomly drawn environments satisfying Assumption 5 with $K \in \{3, 4\}$, convex costs and likelihood-ratio-ordered families, no instance had two or more binding constraints, confirming Proposition 10; among the instances in which some constraint bound, the binding deviation was non-adjacent in 44% of cases. Adjacency is therefore not a generic property of this class, and a model that needs it must assume (19) or its generalisation directly.*

10 Relation to persuasion, expertise, and private information

The closest conceptual comparison is with three strands of information economics.

First, standard Bayesian persuasion studies how a sender chooses an experiment to move a receiver’s common posterior across action regions. Our problem adds an endogenous private input to belief formation: the receiver’s own hidden action determines his prior before he sees the public message. The design object is consequently a collection of linked posteriors rather than one posterior distribution.

Second, Alonso and Câmara (2018) ask whether a sender benefits from becoming an expert—privately observing a signal before selecting a public experiment. Their key issue is whether expertise permits state-contingent experiment selection and whether available public experiments can substitute for that expertise. Our timing and mechanism are different. The principal commits before the hidden action and does not use private expertise to select among experiments *ex post*. Instead, the agent’s privately known action changes how he interprets the outcome of the same committed experiment. The value in our model comes from complementarity between endogenous private history and public feedback, not from adapting the experimental choice to a privately observed sender type.

Third, He, Sandomirskiy, and Tamuz study *private private information*: multiple signals that are informative about a state but statistically independent of one another. Their privacy constraint creates a tradeoff in Blackwell informativeness and a Blackwell-Pareto frontier. Our signal and action history generally need not be statistically independent. Indeed, their correlation through the state is what allows the same public message to have different values across histories. The relevant constraint in our model is incentive compatibility rather than privacy, and optimality is payoff-specific rather than Blackwell-Pareto. Nevertheless, their work clarifies a useful broader perspective: the value of a public signal cannot be assessed in isolation from the private information with which it will be combined.

The notes’ contribution can therefore be stated succinctly:

We study information design when the receiver’s private information is endogenously generated by his own hidden action. A committed public experiment can then serve as an incentive instrument because its decision value depends on that private history. We characterize the resulting multi-constraint persuasion problem by concavification and show, in a spatial binary-action model, that exploiting information complementarity leads to disconnected pooling of extreme low states with relatively high states.

10.1 Comparison with standard persuasion

The distinction can be summarized as follows.

	Standard persuasion	Private-action-history design
Relevant posterior	One common posterior	A linked collection $(\mu, (T_a\mu)_{a \neq a^*})$
Purpose of pooling	Move a common posterior across an action threshold	Make the same message differentially informative across histories
Role of a pooled state	Changes the average belief of one receiver	May barely affect the worker but strongly contaminate the shirker's belief
Incentive channel	Induce a desired continuation action	Reward past effort with greater information value

This mechanism is also distinct from moral-hazard persuasion with a separate receiver. In that class of models, hidden effort changes the distribution of public messages and hence the frequency of receiver actions; the receiver interprets each message using an equilibrium effort conjecture. Here, the same agent both chooses the hidden action and later interprets the signal. His actual private action changes the conditional value of each message and may change the continuation action he takes after that message. The state dependence of the agent's continuation payoff is essential for this channel.

11 Research agenda

Items 1, 2, 4 and 5 are settled in these notes; items 3, 6 and 7 remain.

1. *Settled* (Proposition 2). Strong duality and the linked-posterior concavification result for general finite action sets, with explicit tie-breaking and the support bound $n + r$.
2. *Settled, negatively* (Proposition 10, Example 1, Corollary 2). Under a likelihood-ratio-ordered family with a common diagnostic set, exactly one deviation constraint binds; but it need not be the adjacent one, even with strictly convex costs. Adjacency requires the separate single-crossing condition (19).
3. *Partly settled* (Propositions 6 and 7). Monotonicity of the shadow price, the distortion profile and the pooling cutoff in the effort cost, concavity of the on-path value, and monotonicity in detectability are established for the binary-action spatial model. What remains is the analogous exercise for general K , where the comparative statics must track which deviation is $a^\dagger$ and how $a^\dagger$ itself moves with primitives; the required-depth representation t_a of Section 9 is the natural object.
4. *Settled* (Proposition 5). Under a sharp lower tail the regular-region likelihood-ratio solution is globally valid on the entire implementable range, and $\bar{c}$ is the exact implementability frontier. The linear program (8) handles environments where Assumption 4 fails and the diagnostic set intrudes into the continuation region; that case is open.
5. *Settled* (Corollary 1). The disconnected-pool result, with the sharp criterion (13).
6. *Open*. The repeated-continuation extension with a persistent state. The static linked-posterior operator T_a becomes a family indexed by calendar time, and Bayes plausibility becomes a martingale restriction on the on-path belief process; the substantive question is whether

the incentive contribution h_a retains the difference-of-convex structure of Lemma 1(ii) once continuation values are endogenous.

7. *Open.* Applications to project review, diagnostic systems, education, and human–algorithm decision support.

The central economic statement is:

The principal does not merely design what the agent learns. She designs how useful public feedback is when combined with privately known experience generated by the agent’s own past action.

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